

Other topics

Operator overloading

- Unary Operators (+, -, !, ~, ++, --, true, false)

```
public static Train operator ++(Train wagon)
```

- Binary Operators (+, -, *, /, %, &, |, ^, <<, >>)

```
public static Train operator +(Train wagon, Wagon wagon)
```

```
public static Train operator +(Wagon wagon, Zug zug)
```

```
public static Train operator +(Wagon wagon1, Wagon wagon2)
```

- Comparison operators (==, !=, <, >, <=, >=)

```
public static bool operator ==(Wagon wagon1, Wagon wagon2)
```

Define enumerator for a class

```
public class Train : IEnumerable           //Implement interface
public List<Wagon> Wagons { get; set; }

IEnumerator IEnumerable.GetEnumerator()
{
    foreach(var item in Wagons)
    {
        yield return item;
    }
}
```

```
Train ICE = new Train("ICE");
foreach(Wagon item in ICE)
```

Indexerproperties

```
public class Train
public List<Wagon> Wagons { get; set; }

public Wagon this[int index] //Indexer-Property
{
    get { return this.Wagons[index]; }
    set { this.Wagons[index] = value; }
}
```

```
Train ICE = new Train("ICE");
Wagon w = ICE[3]; //Zugriff erfolgt wie bei Array
```

Timer

- executes an event at a specified time interval
- requires the namespace `System.Timers`;

```
int countdownNumber = 100;
```

```
Timer timer = new Timer();  
timer.Elapsed += new ElapsedEventHandler(Countdown);  
timer.Interval = 1000;  
timer.Start();
```

```
void Countdown(Object sender, ElapsedEventArgs args)  
{  
    countdownNumber--;  
    Console.WriteLine(countdownNumber);  
    if (countdownNumber <= 0)  
    {  
        timer.Stop();  
    }  
}
```